FOOTBALL BUILD SPECIFICATION

Separate repository. Separate database. No shared runtime with the tennis system.

Read `00-developer-brief.md` first. The non-negotiable rules (R1–R10), the arithmetic library, the three measurement layers and the staking rule all apply here and are not repeated.

This is the stronger of the two builds. The core result is peer-reviewed, tested over twelve years and 68,672 bets against a random-betting control, and has not been debunked. It is also thin — around +0.8% per bet — and it has been public since 2020.

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1. Scope

1.1 Leagues

Country	Competitions
England	Premier League, Championship, League One, League Two
Spain	La Liga
Germany	Bundesliga
Italy	Serie A
France	Ligue 1
Scotland	Premiership — **wider threshold, see §7.3**
Europe	Champions League, Europa League
Domestic cups	**Restricted to corners and both-teams-to-score only — see §7.4**

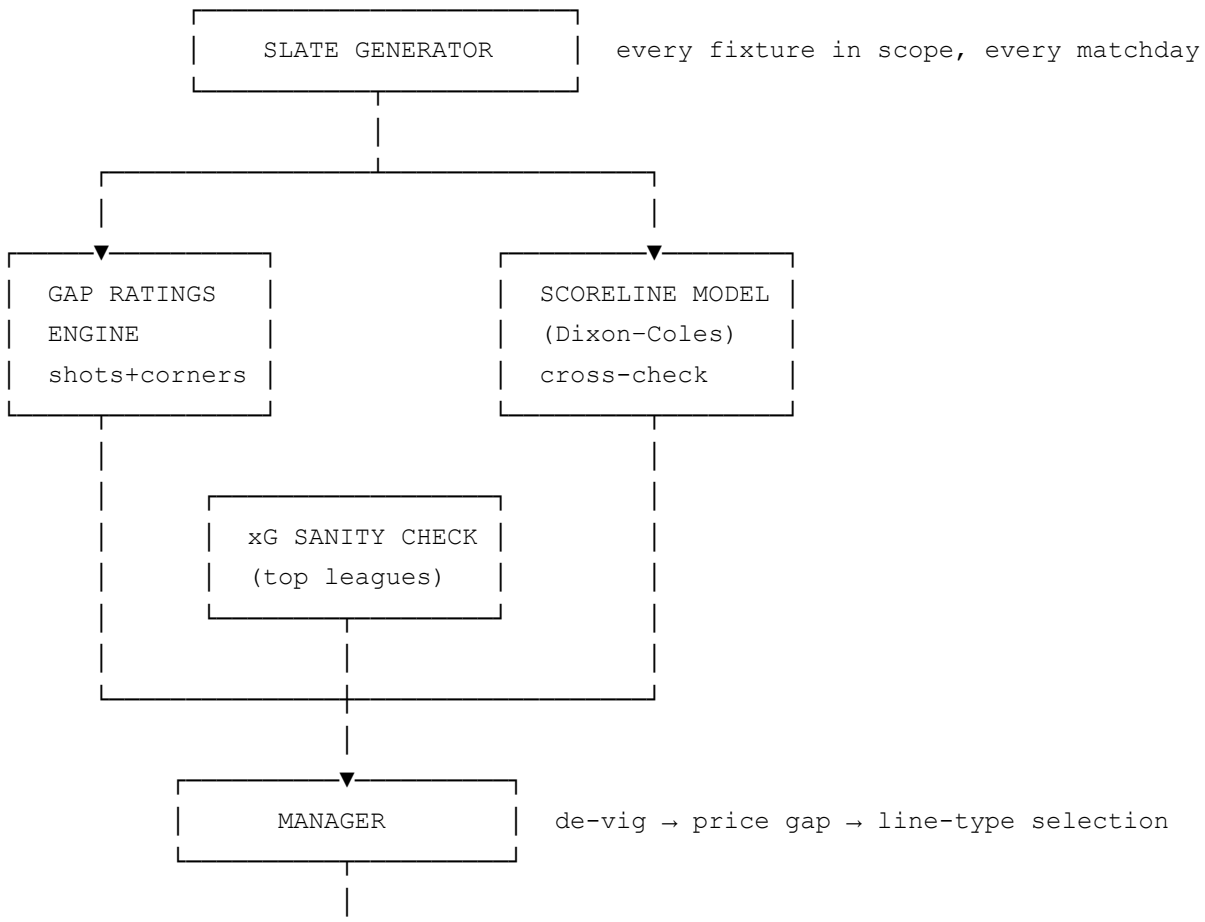
Excluded: the English Conference and below (no reliable public shots, corners or odds data — the GAP engine would run blind); Portugal (thin data, wider margins, untested in the literature); all women's football.

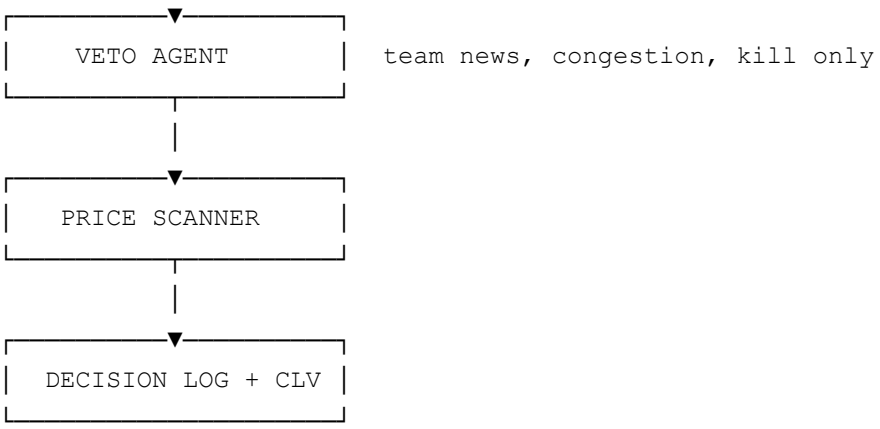
1.2 Markets

Market	Priority	Notes
Over/under total goals	**Primary**	This is the market Wheatcroft's result is measured in
Total corners	High	GAP predicts corners natively; books price them loosely
Both teams to score	High	Robust to rotation, which is why cups stay in
Asian handicap	High	See the line-type rule in §8 — this is a measured edge
Total shots / shots on target	Medium	Same engine, thinner markets
1X2	Low	The most efficient market in football; hardest to beat

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2. Architecture overview





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3. GAP ratings engine — the core

3.1 What it is

Generalised Attacking Performance ratings, from Wheatcroft (2020). Every team carries two ratings per statistic being modelled:

- an **attacking** rating: how much of that statistic the team generates against an average opponent
- a **defensive** rating: how much of it the team concedes to an average opponent

Ratings update after every match against the error between expected and actual.

3.2 Why shots and corners rather than goals

This is rule R1 and it is the whole result. Goals are the noisy outcome of a process; shots and corners are the process. A team that generates twelve shots and eight corners and scores none did roughly the same thing as a team that generated twelve shots and eight corners and scored two — but goals record them as opposites.

Wheatcroft's headline figure — around **+0.8% average profit per bet over twelve years**, validated against a random-betting control — was obtained using **only shots and corners as inputs, not goals**. Substituting goals as an input does not make it a slightly different model, it removes the finding.

Goals may be used for validation. Never as a rating input.

3.3 The update mechanic

For a match between home team `h` and away team `a`, for a given statistic:

```
expected_h = ( attack_h + defence_a ) / 2
expected_a = ( attack_a + defence_h ) / 2
```

That is: what a team is expected to generate is the average of its own attacking rating and the opponent's defensive rating.

After the match, with `actual_h` and `actual_a` observed:

```
attack_h'  = attack_h  + λ · ( actual_h − expected_h )
defence_a' = defence_a + λ · ( actual_h − expected_h )
attack_a'  = attack_a  + λ · ( actual_a − expected_a )
defence_h' = defence_h + λ · ( actual_a − expected_a )
```

λ is the learning rate. Fit it by walk-forward validation, separately per statistic and per league. Home and away ratings should be maintained separately (a team's home attacking output differs systematically from its away output) — the paper handles this; follow it.

Run one independent GAP system per statistic: shots, shots on target, corners. They do not share ratings.

3.4 From ratings to a market price

Wheatcroft's second paper (arXiv 2001.09097) is the important one operationally: it shows the same framework predicting the **match statistics themselves** in advance, and that those predictions carry information beyond what is reflected in the odds. That is what lets one engine price several markets.

Pipeline:

1. GAP ratings → expected shots, shots on target and corners for each team.
2. Expected shots and shots on target → a goal-expectation for each side (a simple parametric conversion — the probability of a shot becoming a goal, fitted per league — is sufficient and is what the paper uses).
3. Goal expectations → a full scoreline distribution (see §4).
4. Scoreline distribution → prices for over/under, both-teams-to-score, Asian handicap, 1X2.
5. Corner expectations → a corner-total distribution directly (negative binomial fits corner counts better than Poisson; test both).

A free Python walkthrough of a GAP implementation exists ("Implementing the GAP ratings in python", on Medium). Useful as a structural reference. Verify it line-by-line against

the paper rather than trusting it — you are responsible for the arithmetic, not the blog author.

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4. Scoreline model — Dixon-Coles

4.1 Purpose

Turn goal expectations into a full 2-D scoreline matrix, from which every goals-based market is derived. This is also the cross-check on the GAP output.

4.2 Structure

```
λ_home = exp( α_home + β_away + γ )
λ_away = exp( α_away + β_home )
```

where α is attack strength, β defence strength and γ home advantage.

Fitted reference values (2017–18 Premier League, from the canonical replication):

- $\gamma \approx 0.2945$ — home advantage as roughly a **1.34× multiplier** on home expected goals.
- $\rho \approx -0.1285$ — the low-score correction parameter.

4.3 The tau correction — do not skip this

Independent Poisson systematically **under-predicts 0-0, 1-0, 0-1 and 1-1**. The Dixon-Coles τ correction adjusts exactly those four scorelines, governed by ρ. Its net effect is to **raise the draw probability**.

Any naive Poisson model will systematically under-price the draw. Setting $\rho = 0$ collapses the model back to plain Poisson, which is a useful unit test.

4.4 Time decay buys very little

$w(t) = \exp(-\xi t)$. The original paper used $\xi = 0.0065$ with half-weeks as the unit. The replication found the optimal ξ is **0 within a single season** and about **0.00325 across five seasons**, for a log-likelihood improvement of 0.23 — essentially nothing.

Read against the grain, this says within-season data should be weighted roughly flat. **Recency weighting is not where the money is.** Do not spend a week tuning it.

4.5 From the matrix to prices

Build `P(i, j)` for `i, j = 0…10`, then:

- **1X2** = lower triangle / diagonal / upper triangle
- **Correct score** = read the cell
- **Over/under N** = sum cells where `i + j > N`
- **Both teams to score** = sum cells where `i ≥ 1` and `j ≥ 1`
- **Asian handicap** = sum cells by `i − j`, splitting quarter-lines into two half-stakes and treating integer lines as a push

Because all of these derive from one object, a mispricing in one implies a mispricing in the others. Check Asian handicap and 1X2 against each other every time; a disagreement is a bug in your derivation, not a discovery.

4.6 Honest expectation

A textbook Dixon-Coles on public data is a **baseline, not an edge**. The practitioner who published the definitive replication said plainly that these models will not make money. It is in the build to cross-check GAP and to price derivatives — not as a source of edge.

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5. xG sanity check

A check, not an input. Do not feed xG into GAP ratings.

Where xG is available (top-five leagues; Understat and FBref cover these, and **not** League One or League Two), compare the GAP-derived attacking assessment against the xG record. If GAP says a team is creating well and xG disagrees materially, flag the fixture and skip it.

5.1 Why it earns its place

Expected goals beats raw goals as a forward predictor: Pearson correlation with future goals of **0.71 versus 0.58**, and next-match result accuracy of **59.2% versus around 50–52%**. Material, but not enormous — which is precisely why a market that also knows this is hard to beat.

5.2 Why it is not an input, and why “xG mean reversion” is not a strategy

The single most-cited edge in football betting is that a team over- or under-performing its xG will regress. Two problems:

1. **The magnitude is unpublished.** I could not find a verifiable figure for how much a team X goals above xG over N games subsequently regresses, in any covered league. The claim is everywhere and anchored to nothing.
2. **Finishing skill is nearly undetectable at bettable sample sizes.** A player genuinely 25% better than average, taking 100 shots a season, has only around a 70% probability of out-performing his cumulative xG in four seasons out of five. Empirical-Bayes work finds essentially zero correlation between a player's shrunk goals-per-xG in one year and the next. Deflections and speculative efforts pollute the measure badly — excluding deflections alone moved one player's goals-above-expectation from 14.61 to 9.03.

Working prior: treat finishing over- and under-performance as noise. And note the deeper problem — the market regresses it too, so even where the effect is real it is not automatically an edge.

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6. Home advantage — get this right, it is widely misreported

The 2020–21 empty-stadium period was a natural experiment. Wunderlich et al., 36,882 matches with crowds against 1,006 without, across ten leagues:

Effect, crowd → no crowd	**Result**
Away-team foul disadvantage	**Eliminated** ($p < .001$)
Yellow-card referee bias	**Disappeared** ($p < .001$)
Red-card bias	**Reversed** ($p < .05$)
Home shot advantage	~50% reduction ($p < .001$)
Home advantage in goals	Decrease **not significant** ($p = .449$)
Home advantage in points	Decrease **not significant** ($p = .498$)

This did **not** show that home advantage is the crowd. It showed a clean dissociation: the crowd drives referee bias, which is real and measurable; the crowd does not drive match outcomes. Residual home advantage — travel, familiarity, tactical asymmetry — survived.

Practical consequence: keep γ fitted on the full history. Do not build a crowd-size feature. And note that referee identity **does** matter for cards markets, which is the one place this result is directly actionable.

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7. The manager

Same principle as tennis: **it votes on the price gap, not on the outcome.**

7.1 Sequence

1. GAP ratings produce expected statistics and, via the scoreline model, a probability for every in-scope market.
2. xG sanity check runs where data exists. Disagreement → skip the fixture.
3. Take the sharp reference price for the market. Compute booksum. **De-vig with power or Shin.**
4. Price scanner supplies `best_available_price`.
5. `edge = (p_model × best_available_price) − 1`
6. Fire if `edge ≥ threshold`.

7.2 Base thresholds

Competition group	Threshold
Premier League, La Liga, Bundesliga, Serie A, Ligue 1, Champions League, Europa League	**5%**
Championship, League One, League Two	**6%**
Scottish Premiership	**8%**
Domestic cups (restricted markets only)	**8%**

7.3 Why Scotland is taxed rather than cut

The instinct to cut it is wrong for the reason the client gave, and I've reflected that here: corners and both-teams-to-score do not care about two-club dominance. A mismatch still produces corners, and a rotated or outclassed side still attacks.

The reason for the higher threshold is **margin, not competitiveness**. Softer leagues carry roughly 6–8% margins against 2.5–4% in the Premier League. Since the model's edge is under 1%, a doubled margin means you need roughly double the edge to earn anything at all, on data that is also sparser. Hence 8%, not exclusion.

The same logic — softer market, but 2–3× the margin, much lower stake limits and worse data — applies to the English lower divisions and is the honest counter to the usual "lower

leagues are soft" claim. There is no academic study of League One or League Two market efficiency; all available claims on it are affiliate-published and at least one contains uncited figures that do not survive inspection.

7.4 Cups — restricted, and why

Early-round domestic cup ties are the hardest fixtures in football to price. Squads are rotated heavily, you do not know the lineup until an hour before, and GAP ratings built on league form may describe a team that does not turn up.

But **the market restriction solves most of this**: corners and both-teams-to-score are far less sensitive to *who* is on the pitch than the match result is.

Rules:

- Cups: **corners and both-teams-to-score only.** No 1X2, no Asian handicap, no correct score.
- Skip any tie involving a club below League Two, where the model has no ratings at all.
- Threshold 8%.

Enforce this as a hard constraint in code, not a convention. Result markets will otherwise creep back in the first time someone adds a market to a config file.

7.5 Fixture congestion flag

Set a flag when a team has played a European fixture in the days before a domestic match, or faces an unusually short turnaround.

Use it as a veto input, not a rating adjustment. Fixture congestion does not do what pundits say it does — the measured effects are much smaller than the narrative — so do not build a congestion coefficient into the ratings. Flag it, let the veto agent weigh it, and log how often it fires so you can measure later whether it is worth anything.

7.6 Minimum acceptable price

```
min_price = (1 + threshold) / p_model
```

Published with every selection. Below it the bet is dead.

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8. Asian handicap line-type selection — a measured, free edge

This is not a model. It is a rule, and it applies to every Asian handicap bet the system produces.

Whelan et al., across 84,230 and 67,373 matches, measured the cost of betting by handicap line type:

Line type	Loss rate
Integer (0, −1, +1, −2…)	**3.24%**
Quarter (.75)	3.57%
Quarter (.25)	3.61%
Half (.5)	**4.16%**

Nearly a full percentage point between the cheapest and most expensive, for structurally similar bets. Against a model edge of under 1%, **the line type you choose matters more than the model that chose the team.**

Rules:

1. Where the same view can be expressed on more than one line, prefer **integer**, then **.75**, then **.25**, and take **.5** only if nothing else is available.
2. The same work found **no favourite-longshot bias in Asian handicap markets**, unlike 1X2 — so unlike the 1X2 market, you do not need to discount longshot AH prices for hidden margin.
3. Record the line type on every pick record so this can be verified in the CLV data later.

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9. The veto agent

Kill only. Never creates a selection, never raises a probability, never relaxes a threshold.

Check	Kills the bet if
Team news	Key absence confirmed after prediction (define “key” by minutes played share, not by fame)
Lineup	Confirmed lineup materially diverges from the expected side
Congestion	Congestion flag set and turnaround below the configured minimum

Weather	Conditions likely to suppress the modelled statistic — heavy rain and high wind for corners and shots markets in particular
Referee	For cards markets only: referee unknown or newly appointed with no record
Postponement	Fixture postponed, moved, or venue changed
Data staleness	Any input feed outside its freshness window

Every veto logged with its trigger.

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10. The price scanner

Identical in design to the tennis scanner (see §8 of `01-tennis-build-spec.md`), and it must produce the same two distinct flag types:

Type A — model edge. GAP plus the scoreline model imply a fair price shorter than the best available by at least the threshold.

Type B — price dispersion. A soft book's price exceeds the de-vigged sharp fair price by at least the threshold, with no reference to our model.

Type B is the better-evidenced. Buchdahl: **+3.40% over 22,281 bets**, rising to **+13.22% over 1,927 bets** where the gap exceeded 5%; control **–3.08% over 44,673 bets**.

Book margins differ enormously and this drives everything: Pinnacle around 2.7% on 1X2, bet365 around 5.5%, Ladbrokes around 7.8%. That spread is several times the size of the model edge.

Corners and cards specifically. These carry *higher* margins than main markets because books are less confident pricing them — on one Premier League over/under 10.5 corners line, the tightest book held just above 7.5% while several rivals were above 10%. So the weak pricing is real, but you pay for the privilege. **Only take corner and card bets at the tightest-margin book available**, and record the booksum on the pick record so this can be audited.

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11. Schedule and publication

Day / time (UK)	Job

Monday	Full week's slate generated, ratings recomputed. **Not published** — no team news yet, and the price may not survive
Daily 06:00	Results ingest, ratings recompute
Continuous	Odds snapshots every 15–30 min for fixtures inside 72 hours
Match day 08:00	Veto agent runs, freshness gate checked, **slate published**
Match day, kick-off minus 3 hours	Final veto sweep; retractions published
Kick-off	Closing price captured automatically

Publishing at 08:00 on the day gives the audience the whole morning, which matters more than the marginally better price available on Monday. The kick-off-minus-three-hours sweep exists because confirmed lineups land about an hour before kick-off, and that is when injuries become facts — so it is a retraction window, not a publication window.

Empty slates are valid output.

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12. Backtest protocol

1. Load the full football-data.co.uk archive — results, shots, shots on target, corners, cards, referee, plus opening and closing odds from around 20 bookmakers, roughly 25 seasons.
2. Walk-forward only. Never fit and test on the same window.
3. **Use closing odds for validation, but opening odds for the strategy.** The strategy should be evaluated at the prices actually available when a selection would have been made — which is when soft books are most beatable, before they have seen the money. Closing odds are the benchmark, not the entry price.
4. Every feature carries `as_of`. The harness refuses to run if any feature post-dates its fixture.
5. **Minimum 300–500 settled selections before discussing any result.**
6. Report calibration first, then CLV, then ROI.
7. **Record the number of variants tested.**

12.1 Expected results

- GAP over/under: roughly **+0.8% per bet** at best, and realistically less. The paper published in 2020, and broad research on published market predictors finds returns decline about 26% out-of-sample and about 58% post-publication — roughly half from data-mining bias and half from the edge being arbitraged once public. Expect meaningfully under 0.8% today. If your backtest shows 5%, you have a bug or a look-ahead leak.
- Dixon-Coles alone: around break-even before margin, losing after it. Correct.
- Type B flags outnumbering Type A. Expected.

12.2 The pattern-finding warning

Winkelmann et al. ran a Monte Carlo on fourteen seasons of a **fully efficient** market and found a **77.63% probability of observing at least one season with statistically significant home bias**. Search seasons, leagues and bias types and you will find "something" four times in five when nothing is there.

Their substantive finding on real data: inefficiencies are short-lived and do not persist across time or leagues, and the number of apparently inefficient periods barely exceeds chance. Bookmakers have improved.

This is why R4 (walk-forward only) and the variant count are not bureaucracy. They are the only things standing between this build and a beautiful backtest that means nothing.